SCHOOL OF COMPUTER SCIENCE

COMP3220
Object-Oriented Software Analysis & Design
Summer 2026 (2265)
COURSE SYLLABUS

LAND ACKNOWLEDGEMENT

The School of Computer Science at the University of Windsor *sits on the Traditional Territory of the Three Fires Confederacy of First Nations*. We acknowledge that this is the beginning of our journey to understanding the Significance *of the history of the Peoples of the Ojibway, the Odawa, and the Pottawatomie.*

INSTRUCTOR:

Dr. Saeed Samet

E-mail: **ssamet@uwindsor.ca**
Office Location: ***5102 Lambton Tower***
Office Hours: **Wednesday, 10:00 am – 11:00 am, MS Teams**

Note: Only email originating from a valid University of Windsor student account will be accepted from students wishing to contact the instructor or use the Bright Space email tool within the course site. Please include your full name, student ID and related course section in your correspondence. Do not spam with multiple or lengthy emails. Should you not receive timely feedback on your inquiries, reach out during office hours directly, or in the event of no response, contact the CS office at ***csinfo@uwindsor.ca*** *for support to access the instructor.*

*The course outline that is available after the end of the second week of the semester will be deemed correct and official.

Never used Microsoft Teams before?
Download the free MS Teams client for your device and login using your UWindsor account (uwinid). There are two ways to reach me, one using the direct chat to Click or tap here to enter text. *and another to our class group if you like to connect with your peers. It is a simple messenger type application allowing you to do chat, voice and video conferences with your prof and fellow students.*
Getting Started - Students | Information Technology Services (uwindsor.ca)

TEACHING ASSISTANT(S):

Please refer to BrightSpace for the GA/TA contact information and updated office hours.

The teaching assistant(s) will hold regular weekly office hours dedicated to helping students. It is highly recommended that you take advantage of this resource by seeking interactive assistance toward understanding the course materials and guidance for completing the homework. Graders can also review your graded work and help correct or fix grading errors.
If you are facing difficulties in the course, please contact the instructor or the teaching assistant(s). You are expected to spend sufficient time completing all the readings and the assigned work.
If you cannot get hold of the teaching assistant(s) during posted office hours or are waiting for a timely response from them, please report the matter promptly to the course instructor with the details of the situation.
If you identify an exceptional assistant who goes above and beyond, please inform the instructor and consider nominating the person for related university/faculty awards for their commitment.

The School of Computer Science provides free tutoring services.
Undergraduate Students https://tutor.myweb.cs.uwindsor.ca/
MAC Students https://tutor.myweb.cs.uwindsor.ca/

PRE-REQUISITES:

COMP-2120 and COMP-2540
No student is allowed to take a course more than two times without permission from the Dean.

LECTURES/LABS:

Online Session/Material: Monday, 7 pm – 9:50 pm, MS Teams

There are no labs for this course.

COURSE DESCRIPTION*:

This course builds on the knowledge of object-oriented programming, data structures, systems programming. Students are introduced to object-oriented software analysis and design concepts (such as cohesion and coupling), and design practices currently used in industry, (such as design patterns and refactoring). These concepts and practices will be discussed through case studies and programming exercises.
**This description is from the official senate-approved calendar*

LEARNING OUTCOMES:

(source: https://ctl2.uwindsor.ca/cuma/public/courses/pdf/cf6e0f97-49ef-4369-8480-19fe639e5ed6**)**
At the end of the course, the successful student will know and be able to:
-- Distinguish the focus of different phases of software development processes and recognize their relationships.
-- Apply industry-standard tools and approaches.
-- Collect and interpret software requirements needed for a large project.
-- Apply and integrate object-oriented design patterns when creating solutions for given problems.
-- Evaluate different design solutions and explain their difference in terms of quality.
-- Integrate changes to and appraise the benefits of properly designed object-oriented software.
-- Appraise the benefits of properly designed object-oriented software.
-- Recognize the benefits of solid software architecture.
-- Select and apply design patterns in object oriented design.
-- Use common terminology, UML diagrams, and design patterns for appropriate communication during software analysis and design.
-- Recognize the complexity of software products and the need of teamwork in software development projects.
-- Appraise object-oriented methodology as one of the software engineering approaches and identify new trends in practice.
Note: Students are strongly encouraged in participating in the course development and update process. Please feel free to make recommendations for changes of the Learning Outcomes, Course Description, and Course Topics to the instructor or the program chair.

REQUIRED TEXTBOOK:

1. Eric Freeman, Elisabeth Robson, **Head First Design Patterns: Building Extensible and Maintainable Object-Oriented Software**, 2021 (Main Textbook)
2. Craig Larman, **Applying UML and patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development**, 3rd Edition, Pearson, 2004

- *Campus Bookstore:* *https://www.bkstr.com/uwindsorstore/home*
- *Leddy Library:* *https://leddy.uwindsor.ca/*

Costs of Textbooks and Other Learning Materials

Each publicly assisted college and university shall ensure that students are informed of the costs of all mandatory textbooks and other learning materials and that they are in the course syllabus for each course.

- Each textbook or other learning material should be individually costed. If the cost for the current year is not available at the time the syllabus is prepared, the most recent cost should be included with a note indicating that it may change; and
- Whether any restrictions would prevent a student from using a second-hand copy of the textbook or other learning material.

Approximate cost of Learning Materials:
1. Head First Design Patterns: Building Extensible and Maintainable Object-Oriented Software (Paperback 83 CAD)
2. Applying UML and patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development (Hardcover: 96.80 CAD or Paperback 43.23 CAD)

COURSE EVALUATION:

Click or tap here to enter text.

Assignment	15%	3 Assignments
Quiz	15%	10 Online Quizzes
Midterm Exam	**30%**	**Friday, June 26, 2026**
Final Exam	40%	TBA

COURSE SCHEDULE:

Topics*

(The instructor reserves the right to change the outline to accommodate student pace and understanding of the subject matter.)

- Overview of OO programming and UML
- OO Design Patterns with case studies
  1. Introduction to Design Pattern
  2. Observer Pattern

3. Decorator Pattern
4. Factory Pattern
5. Singleton Pattern
6. Command Pattern
7. Adapter and Facade Pattern
8. Template Method Pattern
9. Iterator and Composite Pattern
10. State Pattern

****Note:** Students are advised that the schedule and topics described above are tentative and that the material and/or depth and order of presentation are subject to change at the discretion of the instructor and student pace.*
This course assumes the student will allocate a significant amount of independent study and time spent reading and researching materials as needed. You are strongly encouraged to ensure sufficient time to succeed in this course.

IMPORTANT DATES:

Winter 2026

Monday May 4: First day of classes
Sunday May 10: Academic Add/Drop for Inter-Session 2026 (6 weeks)
Sunday May 17: Academic Add/Drop for Full Summer 2026 (12 weeks)
Monday May 18: Victoria Day University CLOSED
Monday June 1: Last Day to Voluntarily Withdraw from Inter-Session 2026 (6 week)
Saturday June 13: Reading Week for Full Summer 2026 (12 week) courses: June 13-21
Monday June 15: Last Day of Inter-Session 2026 (6 weeks) Classes and Make-up Date for Victoria Day Holiday Classes
June 18: Final Exams for Inter-Session 2026 (6 weeks): June 18-20
Monday June 22: First Day of Classes: Summer 2026 Session (6 weeks)
Sunday June 28: Academic Add/Drop for Summer 2026 (6weeks)
Wednesday July 1: Canada Day: University CLOSED
Tuesday July 14: Last Day to Voluntarily Withdraw for Full Summer Session 2026 (6 weeks)
Monday July 20: Last Day to Voluntarily Withdraw from Summer Session 2026 (6 weeks)
Monday August 3: Civic Holiday: University CLOSED
Tuesday August 4: Make-up date for Canada Day Holiday Classes
Wednesday August 5: Last Day of Full Summer 2026 (12 weeks) Classes and Make-up date for Victoria Day Holiday Classes
Saturday August 8-19: Final Exams for Full Summer 2026 (12 weeks) and Summer 6 (weeks) Session: Aug 8-19 (excl Sundays)
Thursday August 20: Alternate Final Exam Day
Monday September 7: Labour Day: University CLOSED

RESOURCES:

The course website is ***https://brightspace.uwindsor.ca/***
Please check it frequently for announcements and other useful info.

GRADING:

A numeric grade on a scale of 0 to 100 will be assigned (rounded integer).

Passing grade:
A minimum grade of 50% is required to pass this course (70% for grad courses). Your individual program may have higher requirements to maintain good standing; please consult your program requirements and plan accordingly. If you are registered in a course and do not attend or participate or write any evaluations will be assigned a grade of NR (No report). You must withdraw from the course if you do not wish to attend it; not showing up does not constitute withdrawal and will impact your academic record.

Voluntary withdrawal (dropping the course):
You may drop a course within the first 2 weeks add/drop period (1 week in case of 6-week courses) without it showing up on your academic record. Please check with the Registrar’s office calendar on the important dates for withdrawing

voluntarily from a course after the add/drop period should you feel you need to withdraw. It is strongly recommended that you seek academic advice from your instructor or an academic advisor prior to withdrawing from courses.

Absences due to medical or other extenuating circumstances:
Medical leaves, illness, death (in the family), and other difficult circumstances as determined in bylaw 54 are at times unavoidable and would interrupt your academic career. You must report any issues to the instructor as soon as possible prior to considering any academic accommodations. The instructor reserves the right to determine if an accommodation is merited and if the nature of the accommodation is related to the course evaluation. All requests for alternate considerations on medical grounds or other difficult matters must be made in writing (email) to the instructor along with supporting documents before the end of the course. No alternate accommodations will be considered after the end of the course.

Makeup and missed assessment policy:
If you miss a test, assignment, or other assessment in the course, you will receive a zero mark for the missed work. If you wish to have alternate considerations for a valid reason (as per Senate bylaw 54), you must inform the instructor in writing (email) as soon as possible, preferably before the assessment, and not later than seven calendar days. Considerations for any make-up or late submissions will be done on a case-by-case basis on compassionate grounds while maintaining fairness as much as possible. Alternate considerations will only be given to any missed assessment if the instructor is informed within seven calendar days after its due date. The instructor will refuse any unsubstantiated and late requests.

There is no make-up for the midterm exam. In documented exceptional cases (as per the Senate Bylaws), the weight of missed midterm exam may be partially added to the final exam.

Grade appeal:
Informal reviews and appeals of the marks for assignments, midterm, exams and/or projects will be considered only if requested within 10 days after the release of the corresponding grades. After the 10 days, students will have to submit a formal appeal if they wish within 6 weeks. See Senate Bylaws 54 (Undergraduate Students) and Senate Bylaws 55 (Graduate Students) for more details on appealing grades.

Other Notes:
1. A. Undergraduate Students: (Please review Bylaw 54) The last seven calendar days prior to and including the last day of classes are free from any procedures for which a mark will be assigned. (Extensions on compassionate grounds are excluded). (In the case of six-week courses, the last three calendar days before the start of the examination period are free from any assessment procedures).
1.B. Unannounced quizzes/graded activities will be at most 5% of the final grade.
1. C. Participation marks in online courses will be at most 20% of the final grade.

2. The final exam schedule is announced by the Registrar's office, generally after the add/drop period, and students are expected to be available for the entire exam period and not make any prior travel plans, vacations, or other commitments until after the exam dates are announced. No alternate exam accommodations will be made on those grounds.

3. No forms of assessment shall be scheduled or made due on days identified as break days, such as reading weeks, holidays, or days that the University is officially closed.

SPTs:

The Student Perceptions of Teaching (SPTs) forms will be administered in the last two weeks of classes for courses 12-24 weeks in duration, in the last week of classes for courses 6-11 weeks in duration, or in the last two days of classes for courses of 5 or fewer weeks in duration. Students should be provided with up to 15 minutes at the beginning of a class to complete the SPTs online. Senate Policy

SUPPORT CONTACTS:

The School of Computer Science has a team of support staff and access to student academic advisors to assist you with any inquiries you may have about our courses and programs. Please use one of the following emails:
For CompSci undergraduate programs and advising, including IT certificate: csinfo@uwindsor.ca
For CS Tutors (free tutoring support for all CS undergrad courses): http://tutor.cs.uwindsor.ca/
For Computer Science Society: https://css.uwindsor.ca/
For CompSci graduate programs (MSc, MSc-AI stream, and PhD): csgradinfo@uwindsor.ca
For CompSci professional graduate programs (MAC/MAC-AI stream): macprogram@uwindsor.ca
For the office of the Director of the School of Computer Science: csdir@uwindsor.ca
For CompSci technical support: https://help.cs.uwindsor.ca/
For International Student Centre: https://www.uwindsor.ca/international-student-centre/
For Student Accessibility Services: https://www.uwindsor.ca/studentaccessibility/
For other general inquiries, https://ask.uwindsor.ca/
For Student counselling services (ext. 4616): https://www.uwindsor.ca/studentcounselling/
For Student health services (ext. 7002): https://www.uwindsor.ca/studenthealthservices/

For Student Peer Support Centre (ext. 4551): https://www.uwindsor.ca/studentexperience/wellness/
For USci Faculty of Science student support network: https://www.uwindsor.ca/science/usci/

Good2Talk provides free, 24/7, single-session professional counselling and referral by phone to post-secondary students in Ontario. Services are provided in English and French, with translation services available in 100+ languages.

- Call: 1-866-925-5454 (reach professional counsellors)
- Text: GOOD2TALKON to 686868 (reach trained volunteers)

STUDENT ACCOMMODATIONS:

Students with disability:
Students who require academic accommodations in this course due to a documented disability must contact an Advisor in Student Accessibility Services (SAS) to complete SAS Registration and receive the necessary Letters of Accommodation. After registering with SAS, you must present your Letter of Accommodation and discuss your needs with the course instructor as early in the term as possible. Please note that deadlines for the submission of documentation and completed forms to SAS are available on their website:

- http://www.uwindsor.ca/studentaccessibility/

Exam conflicts:
If you have a conflict with two exams at the same time, you will need to talk to both instructors and ask which one is willing to move your exam to a different day or time.

If you have a conflict with examinations due to the following reasons, view the Office of Registrar Alternative Final Exam Policy:

- Conflict with religious conviction during the regularly scheduled time slot.
- Three or more final examinations in a 24-hour period.

Religious Observances:
Requests for accommodation of specific religious or spiritual observance must be presented to the instructor no later than 2 weeks prior to the conflict in question (in the case of final examinations within two weeks of the release of the examination schedule). In extenuating circumstances, this deadline may be extended. If the dates are not known well in advance because they are linked to other conditions, requests should be submitted as soon as possible in advance of the required observance. Timely requests will prevent difficulties in arranging constructive accommodations.
religious_accommodation_for_students.01mar2013.web_ver.pdf (uwindsor.ca)

PRIVACY AND COPYRIGHTS:

Content confidentiality:
Lectures, examinations, quizzes, assignments, and projects given in this course are protected by copyright. Reproduction or dissemination of examinations or the contents or format of examinations/quizzes in any manner whatsoever (e.g., sharing content with other students or websites), without the express permission of the instructor is strictly prohibited. Students who violate this rule or engage in any other form of academic dishonesty will be subject to disciplinary action under Senate Bylaw 31: Student Affairs and Integrity.

Recording of lectures:
Lectures and discussions can be recorded by requesting explicit permission from the instructor. Students planning to do so shall send a request (via email is sufficient) before the lecture is delivered. Students, however, are not allowed to post or share any recorded material to any other individual or party outside of this course.
See Senate Policy on recording lectures.

SAFETY, ACADEMIC INTEGRITY, AND NON-ACADEMIC MISCONDUCT:

Equity, Diversity, and Inclusiveness (EDI)
This course, along with all its components such as lab sections are, without question, safe places for students of all races, genders, sexes, ages, sexual orientations, religions, disabilities, and socioeconomic statuses. Disrespectful attitude, sarcastic comments, offensive language, or language that could be translated as offensive and/or marginalize anyone are absolutely unacceptable. Immediate actions will be taken by the instructor to protect the safety and comfort of the students. An ethnically rich and diverse multi-cultural world should be celebrated in the classroom. The instructor, too, must treat every student equally and with the respect and compassion that all students deserve. Furthermore, UWindsor is committed to combatting sexual misconduct. All members are required to report any instances of sexual misconduct, including harassment and sexual violence, to the Sexual Misconduct Response & Prevention Office so that the victim may be provided with appropriate resources and support options.

- https://www.uwindsor.ca/sexual-assault/
- For police/ambulance emergencies, call 911 (in Canada)
- For campus police, call 519-253-3000 ext. 4444 for emergency and 1234 for non-emergency issues.

Academic Integrity
Please refer to: https://www.uwindsor.ca/academic-integrity/

As defined in the University of Windsor's **Student Code of Conduct**, plagiarism is the act of copying, reproducing or paraphrasing significant portions of one's own work, or someone else's published or unpublished material (from any source, including the internet), without proper acknowledgement, representing these as new or as one's own.

Tips and resources to help you prevent plagiarism:
https://www.uwindsor.ca/academic-integrity/sites/uwindsor.ca.academic-integrity/files/tips_for_preventing_plagiarism.pdf

The instructor will put a great deal of effort into helping students to understand and learn the material in the course. However, the instructor will not tolerate any form of cheating. The instructor will report any suspicion of academic integrity to the Director of the School of Computer Science. If sufficient evidence is available, the Director will begin a formal process according to the University Senate Bylaws which will lead to more review, a strict punishment if convicted, and a note on your permanent student record.

The following behaviours will be regarded as cheating:

- *Copying assignments or quizzes or presenting someone else's work as your own.*
- *Allowing another student to copy an assignment/project from you and present it as their own work; protect your own work and never share it with anyone!*
- *Copying from another student or any other unauthorized source during a test or exam.*
- *Falsifying your identity during the exam or having someone else assist or complete your assessment.*
- *Referring to notes, textbooks, and any unauthorized sources during a test or exam (unless otherwise stated).*
- *Speaking or communicating without permission during a test or exam.*
- *Not sitting in the pre-assigned seat during a test or exam.*
- *Communicating with another student in any way during a test or exam.*
- *Having unauthorized access to the exam/test paper prior to the exam/test.*
- *Explicitly asking a proctor for the answer to a question during an exam/test.*
- *Modifying answers after they have been marked.*
- *Any other behaviour which attempts unfairly to give you some advantage over other students during the grade-assessment process.*
- *Refusing to obey the instructions of the officer in charge of an examination.*

The list given above is not exhaustive. More examples are given in Appendix A, Senate Bylaws 31 – Complete guidelines and procedures on the sanctions imposed by the university are also listed in Table A.1 of the Senate Bylaws 31

In this course any assessment that is deemed plagiarized or in violation of the academic integrity policy will NOT BE GRADED and receive a grade of ZERO unless a different ruling is provided by the adjudication committee formally reviewing the case.

Examples of sanctioning include: *(from Table A.1 in Appendix A of Bylaw 31)*
For first offence: mark reduction up to zero, censure 6-12 months; and for subsequent offence: suspension 4-24 months, censure up until graduation.

Plagiarism detection software:
Plagiarism-detection software *TurnItIn* will be used for all student assignments in this course. You will be advised how to submit your assignments. Note that students' assignments that are submitted to the plagiarism-detection software become part of the institutional database. This assists in protecting your intellectual property. However, you also have the right to request that your assignment(s) not be run through the student assignments database. If you choose to do so, that request must be communicated to the course instructor in writing at the beginning of the course. The instructor reserves the right to choose another plagiarism detection software and students will be notified once it is used.

Use of Generative AI (Artificial Intelligence) tools is prohibited:
In this course, use of any generative AI system (including, but not limited to ChatGPT, Claude, Jenni, Github Co-pilot, DaLL-E, and Midjourney) is considered an unauthorized aid that may provide an unearned advantage, and therefore may not be used in the creation of work submitted for grades or as part of any assignment in this class. Use of generative AI systems in graded assignments for this course is considered academic misconduct and may be subject to discipline under Bylaw 31: Academic Integrity.